

GCC plugin for static analyzer support

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Motivation

- Modernize legacy architecture.
- Support more possible analyzers.

Goals

- A modern C++ alternative.
- JSON export and built-in visualization.
- Backward compatibility with Predator.
- Test the solution across GCC versions.

Figure 1: Modernization of the legacy architecture

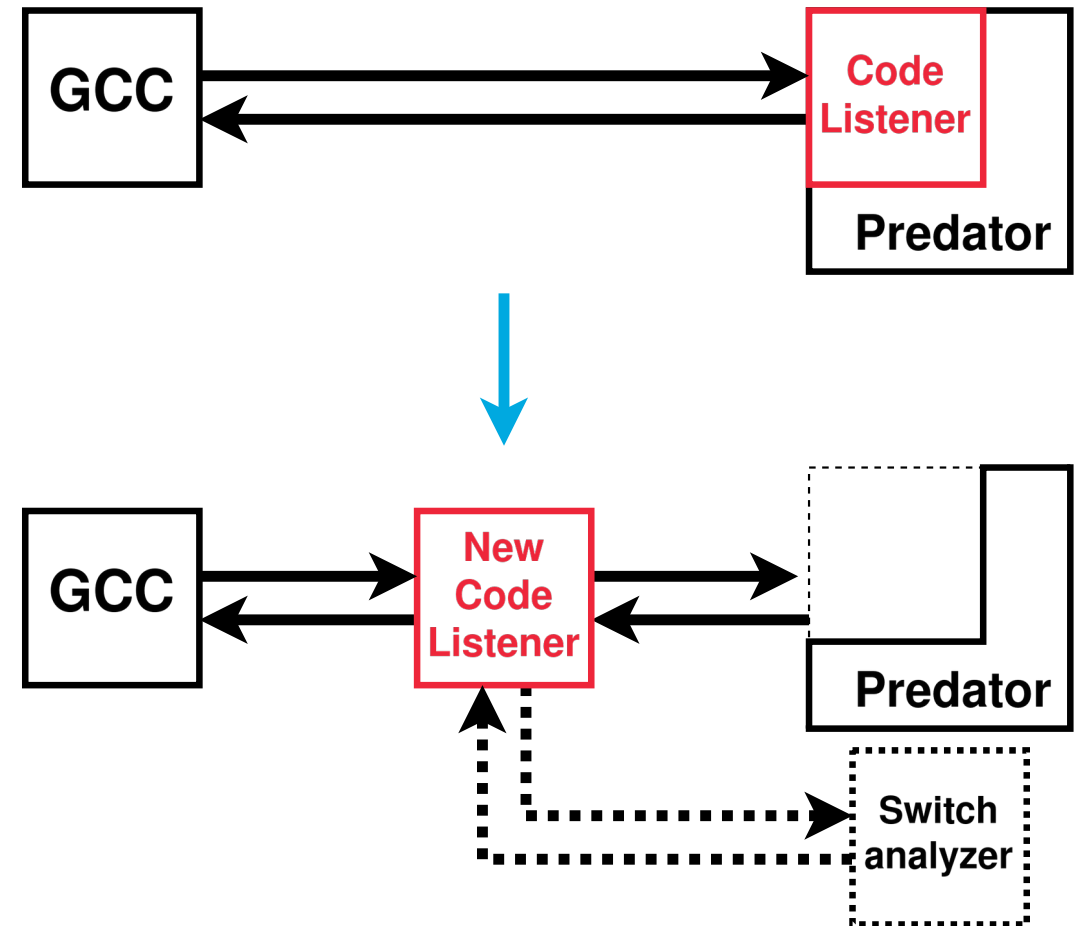


Figure 2: Current Code Listener architecture

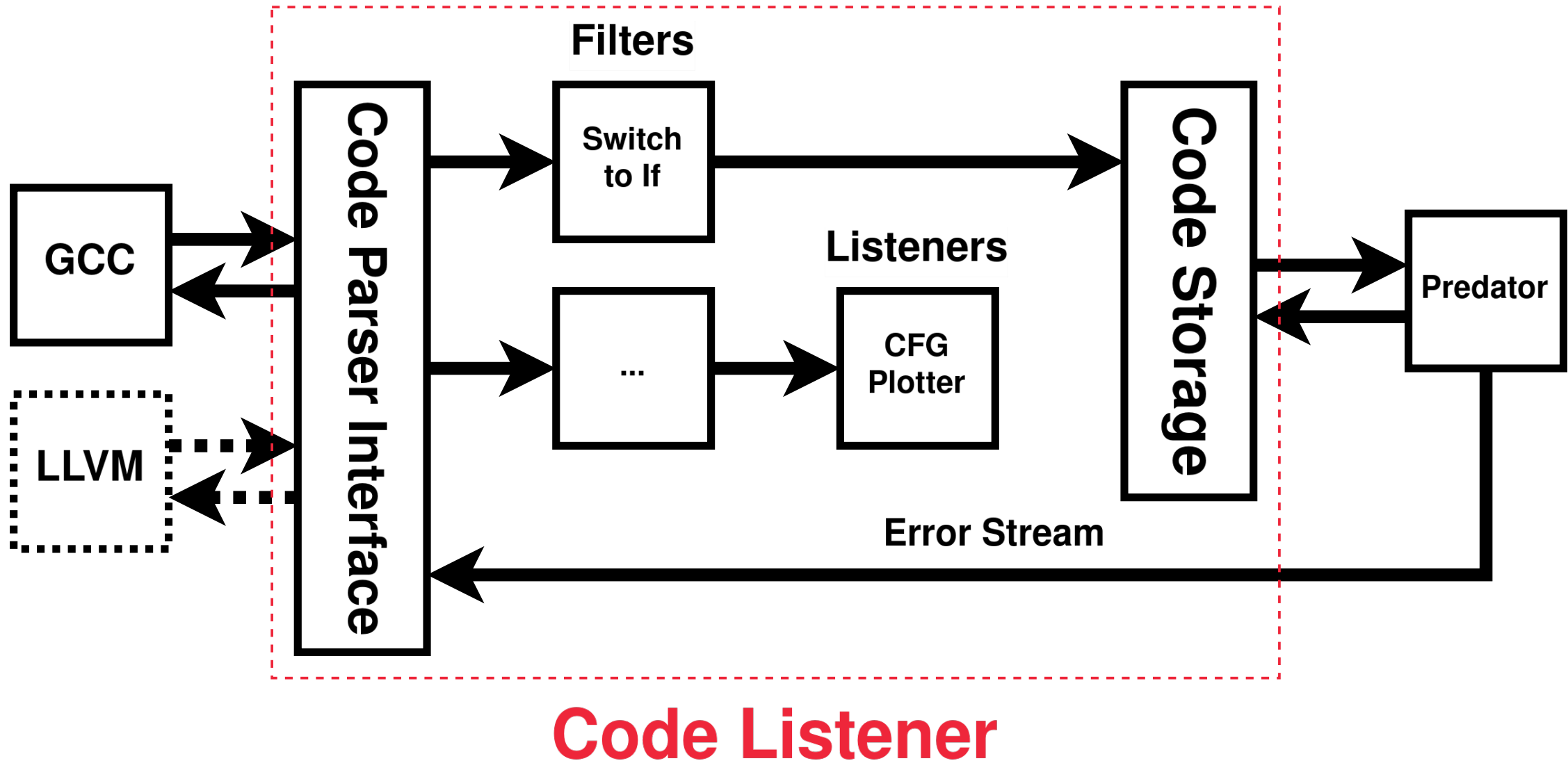


Figure 3: High level architecture of the new plugin

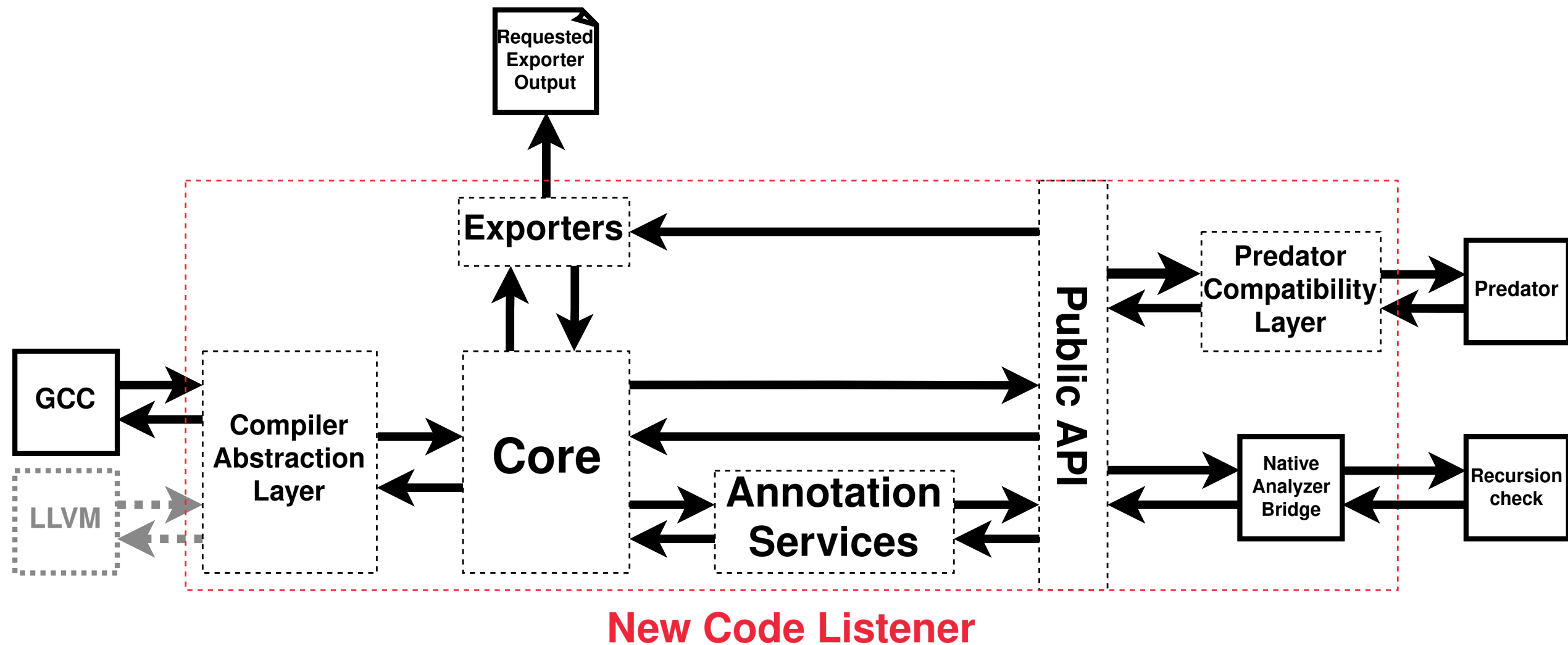


Figure 4: Detailed view of the CAL, Core and Exporters

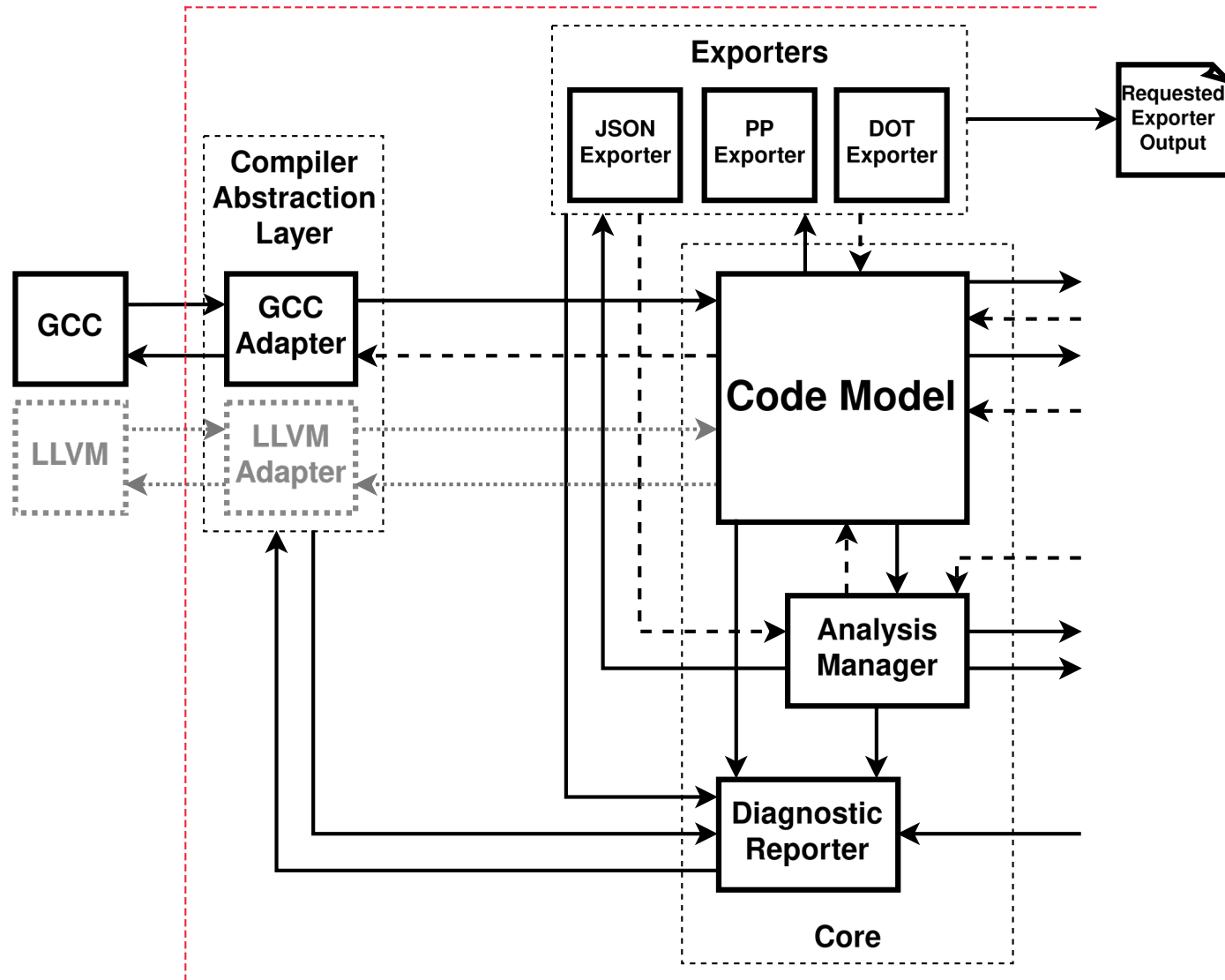
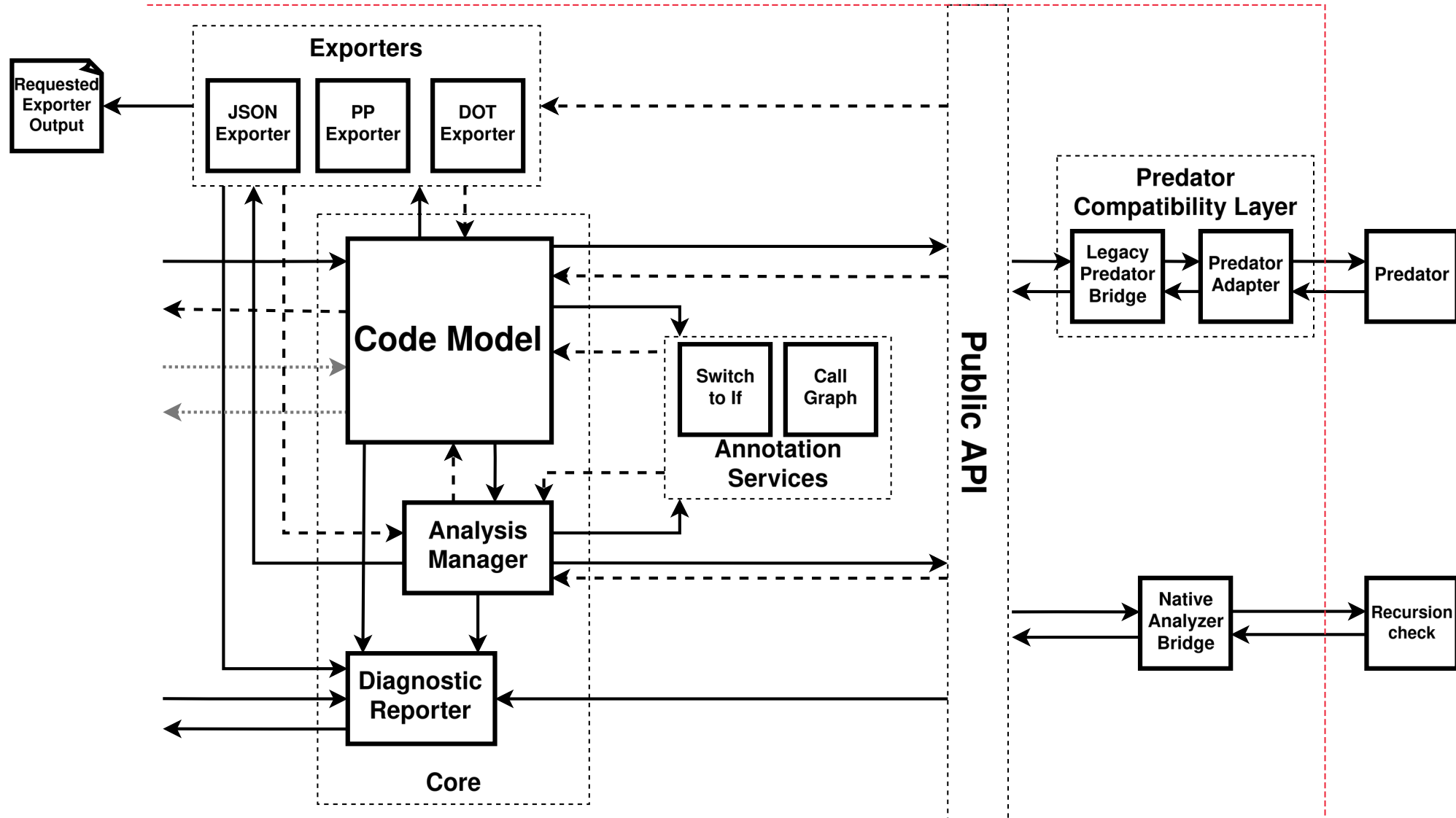


Figure 5: Detailed view of the Exporters, Core, Annotation Services and PCL



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```
auto &cg =  
ctx.analysis_manager.GetAnnotation  
<CallGraph>(m);
```

}

}

Figure 6: cl_merge and cl_analyze alternative pipeline

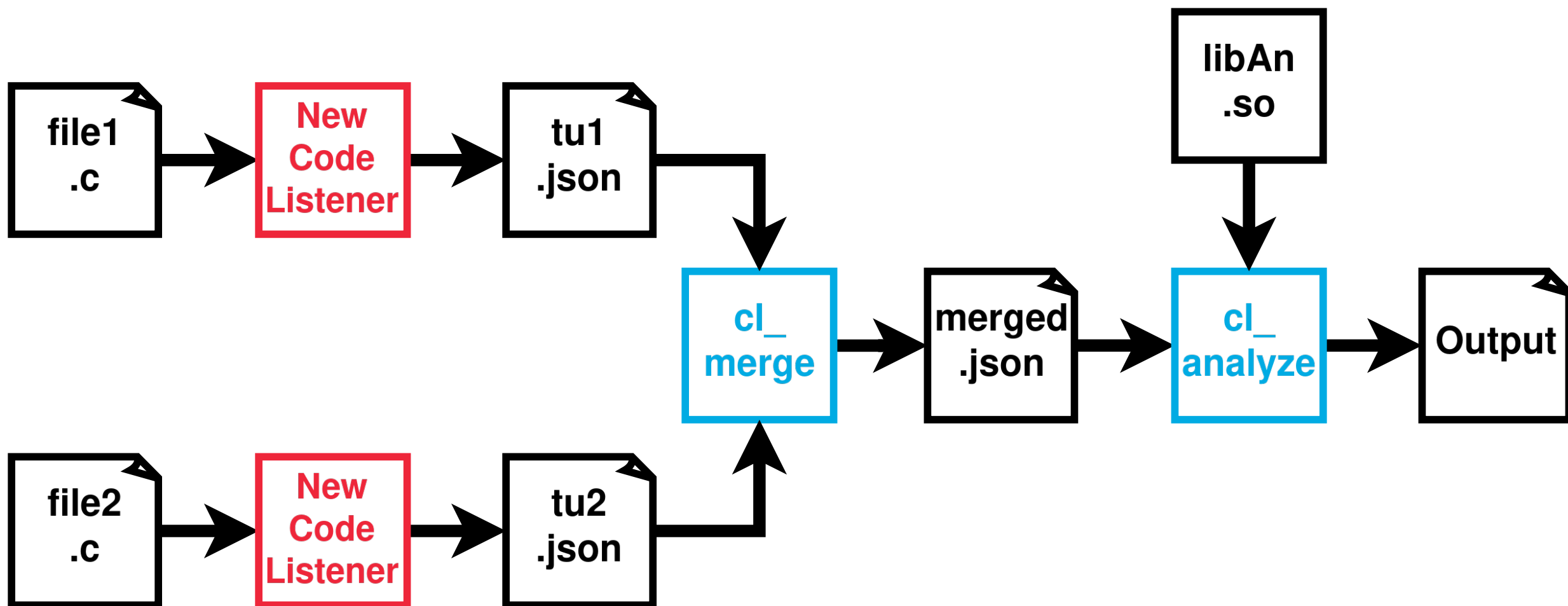


Figure 7: Automated testing via CTest

Test Suite	Number of Tests	Execution Time
Predator Regression	1192	11.52s
GCC Adapter C	46	0.37s
Code Model	38	0.26s
Offline Workflow	3	0.04s

- *Success Rate:* All **1279** tests passed (**100%**) across GCC 12, 13, 14 and 15.
- *Total Time:* End-to-end analysis time **matches** the legacy implementation (~306.8 s vs ~307.1 s).

Contributions

- Modernized C++ architecture.
- Built-in JSON export and visualization.
- Offline whole-program analysis.
- Backward compatibility with Predator.

Future Work

- *Alternative Frontends*: LLVM/Clang support.
- *New Analyzers*: Next-generation Predator.
- *More Exporters*: BroomExporter.

“Analyzátor Broom <https://pajda.fit.vutbr.cz/rogalew/broom> načítá vstup z JSON formátu vytvořeného původním Code-listenerem. Je formát generovaný Vaším nástrojem kompatibilní s původním JSONem generovaným Code-listenerem?”

— doc. Mgr. Adam Rogalewicz, Ph.D.

Feature	Code Listener	New Code Listener
Data Format	Yojson (OCaml tuple, variant)	JSON (RFC 8259)
Hierarchy	Nested	Flat entity pools linked by IDs
Entry/Exit Blocks	Not present	Explicit
Type Size Measured in	Bytes	Bits

Requirements

Dependency	Version	Notes
GCC	12+	Auto-detected by default; can also be forced with <code>TARGET_GCC</code>

3. Configure

```
make configure
# or
cmake -B build -DTARGET_GCC=gcc-12
```



Optional CMake variables:

Variable	Default	Description
<code>TARGET_GCC</code>	auto-detected	GCC executable name or absolute path